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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

A Project Phase-1 Report

On

“HOME SECURITY SYSTEM USING ESP32-CAM”

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CERTIFICATE

Certified that project phase-I work entitled “**Home Security System Using ESP32-CAM**” carried out by, **Mr. Amit Dhavaleshwar (2NR22AD002)**, **Mr. Anil Kumar R (2NR22AD004)**, **Ms. Hema Chalesara (2NR22AD009)**, **Mr. Vishnu Shetty (2NR22AD039)** are Bonafide students of VI semester in partial fulfilment for award of **Bachelor of Engineering in Artificial Intelligence and Data Science** of Visvesvaraya Technological University, Belagavi during the year 2024-25. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in report deposited in departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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ABSTRACT

Advanced IoT-based Home Security and Surveillance System designed using the ESP32-CAM microcontroller, integrating artificial intelligence, real-time monitoring, and smart home automation principles. The system functions as a comprehensive, low-cost, and scalable security platform capable of detecting intrusions, capturing visual evidence, and alerting homeowners instantly through network-enabled communication.

The proposed model incorporates a combination of PIR motion sensors, ESP32-CAM-based image acquisition, Wi-Fi connectivity, and optional facial recognition modules to provide an intelligent, automated security workflow. Upon detecting human presence, the system activates image/video capture, processes the data, and transmits alerts to the user through a mobile device or web interface. This ensures rapid threat detection while minimizing false alarms through optimized sensor-camera coordination.

The platform adopts a modular architecture that supports future enhancements, including cloud storage, advanced computer vision models, and seamless integration with existing smart home devices. A user-friendly control interface enables homeowners to manage, monitor, and customize system operations remotely, ensuring both security and convenience.

Furthermore, the system demonstrates how IoT-enabled embedded platforms can bridge the gap between traditional security practices and modern intelligent surveillance requirements. By combining autonomous sensing, real-time data transmission, and adaptive monitoring, the ESP32-CAM-based security framework establishes a reliable foundation for next-generation smart home ecosystems. The project highlights not only the technical feasibility of low-power, networked security devices but also their potential to provide scalable, user-centric, and future-ready protection solutions suitable for a wide range of residential and commercial environments..

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CHAPTER I

INTRODUCTION

1.1 OVERVIEW

In today's rapidly evolving digital world, ensuring the security of residential spaces has become a critical necessity. With the increasing rate of home intrusions, thefts, and unauthorized access attempts, conventional security systems often dependent on manual monitoring or basic alarm mechanisms are no longer sufficient to provide reliable protection. These traditional systems lack intelligence, adaptability, and real-time decision-making capabilities, resulting in delayed responses, false alarms, and limited owner control. The emergence of Internet of Things (IoT) technologies has revolutionized modern security solutions by enabling smart, interconnected devices capable of sensing, processing, and responding autonomously.

Project aims to design and implement an AI-assisted Home Security System using ESP32-CAM, integrating hardware sensing technologies with software-driven automation to deliver a comprehensive security framework. The system combines motion detection, image capturing, remote surveillance, and real-time alerts, enabling homeowners to monitor their premises from anywhere using internet-enabled devices. Unlike conventional systems that automatically grant access upon detection, this platform ensures owner verification, significantly enhancing security and reducing unauthorized entry risks. Rapid advancements in computer vision, IoT connectivity, and embedded systems have enabled the creation of smart home environments where multiple devices cooperate to ensure safety, speed, and user convenience

In addition, the system integrates modern communication features that enhance user control and awareness. Real-time alerts are sent directly to the homeowner, allowing quick evaluation of any detected activity. The user can then manually operate the door mechanism through a Bluetooth-based mobile application. This combination of automation and human decision-making provides a highly secure and flexible home security solution.

1.2 APPLICATIONS

1. **Residential Security:** The system provides continuous monitoring of the home environment using motion detection and camera-based verification. It sends real-time alerts to homeowners, enabling quick responses during suspicious activities. Captured images or videos help confirm intrusions instantly. This greatly enhances the safety and protection of family and property.
2. **Commercial / Shop Surveillance:** Shops and small businesses can use the system for round-the-clock surveillance. It detects unauthorized entry after business hours and notifies the owner immediately. The ESP32-CAM captures visual evidence for verification and record-keeping. This reduces theft, improves safety, and supports better shop management.
3. **Remote Property Monitoring** Owners can monitor distant locations such as farms, warehouses, or rental properties in real time. The system sends images, alerts, or live streams through the internet for remote visibility. This ensures continuous awareness even when the owner is far away. It helps prevent trespassing, damage, or unnoticed incidents
4. **Smart Home Integration:** The system can be connected with other IoT devices like alarms, smart locks, or lights. It enables automated, intelligent responses during security events. Users can control or configure the system from their smartphone or home network. This makes it a valuable component of modern smart home ecosystems. •

1.3 ADVANTAGES AND DISADVANTAGES

ADVANTAGES

1. **Low Cost and High Efficiency:** The ESP32-CAM provides powerful processing and camera capabilities at a very affordable cost. This makes the system accessible for most homeowners without compromising performance. Its sensors and modules consume minimal power while offering reliable surveillance. Overall, it delivers professional-level security at a fraction of traditional system costs
2. **Real-Time Alerts and Monitoring:** The system sends instant notifications when motion or intrusion is detected. Homeowners can view live images or streams from anywhere through Wi-Fi. This rapid communication greatly improves response time during security threats. It ensures users remain aware of activities around their property at all times.

3. **Scalable and Customizable:** The modular design allows integration of additional features like facial recognition or cloud storage. Users can expand or modify the system based on their needs and environment. Different sensors or automation components can easily be added. This flexibility makes it suitable for both small homes and larger setups.
4. **Reduced False Alarms:** The combined use of PIR sensors and camera verification improves accuracy. This prevents unnecessary alerts caused by pets, shadows, or environmental changes. The camera provides clear visual confirmation before notifying the user. This enhances trust and reliability in the overall security system.
5. **Energy Efficient Operation:** ESP32-CAM and PIR sensors consume very little power during standby mode. The camera activates only when motion is detected, conserving energy. This makes the system suitable for continuous 24/7 operation. Low power usage also supports battery-based or solar-powered installations.

DISADVANTAGES

1. **Internet Dependency:** The system relies heavily on stable Wi-Fi or internet connectivity. Poor or unstable networks can delay alerts or affect live streaming. This may reduce performance during critical moments. Therefore, the effectiveness depends on network availability.
2. **Limited Night Vision Capability:** The ESP32-CAM requires additional IR LEDs for proper low-light performance. Without them, image clarity may decrease at night. This can affect intrusion detection in dark environments.
3. **Processing Limitations:** The ESP32-CAM cannot run heavy AI or advanced detection algorithms locally. Complex tasks require cloud services or an external processor. This limits its ability to perform detailed analysis on-device. Therefore, expansion may require additional computational components.
4. **Sensor and Coverage Constraints:** PIR sensors have limited detection angles and distances. Incorrect placement can reduce accuracy or miss certain areas. Environmental factors may influence sensor sensitivity. Thus, proper setup and alignment are crucial for optimal performance.

CHAPTER II

PROBLEM STATEMENT AND OBJECTIVES

2.1 PROBLEM STATEMENT

Traditional home security systems rely on basic alarms or manual monitoring, which often results in delayed responses, false alerts, and limited user control. Many smart systems lack proper owner verification and can unintentionally grant access, creating security risks. There is a need for a low-cost, intelligent, and remotely accessible IoT-based solution that can accurately detect intrusions, capture real-time evidence, and immediately notify homeowners. An ESP32-CAM-based system can address these limitations by combining motion detection, image acquisition, and wireless communication to provide reliable and efficient home surveillance.

2.2 OBJECTIVES

1. To develop an IoT-enabled home security system using ESP32-CAM that can detect intrusions and capture real-time visual evidence through automated image or video acquisition.
2. To design an intelligent monitoring framework that integrates PIR sensors, onboard camera modules, and wireless communication to ensure accurate threat detection with minimal false triggers.
3. To enable real-time remote surveillance by transmitting alerts, images, and live-streaming data to the homeowner's connected device via Wi-Fi or cloud-based services.
4. To ensure enhanced user control by allowing homeowners to verify intrusions manually instead of relying on automatic access control systems, thereby reducing unauthorized entry risks.
5. To create a scalable and cost-efficient solution that can be easily expanded to include features like facial recognition, cloud storage, smart locks, and advanced computer vision algorithms.
6. To provide a user-friendly interface that simplifies system monitoring, configuration, and control, ensuring accessibility even for non-technical users.

CHAPTER III

LITERATURE SURVEY

TITLE 1: Smart Mirror Using ESP32 for Intrusion Detection**Authors:** T. B. R. Jeyanthi, R. Nivetha**Publish Year:** 2019

About: This study proposes a smart mirror integrated with an ESP32 microcontroller, PIR sensor, and embedded camera to detect human presence and send real-time alerts. The authors focus on creating a dual-purpose device—functioning as a household mirror while serving as a security module. Motion detected by the PIR sensor triggers the ESP32 camera to capture images and notify the user through Wi-Fi communication. The paper emphasizes low-cost implementation and user-friendly design. However, the model has limited image-processing capability and may generate false alarms in complex environments, indicating the need for more intelligent verification techniques.

TITLE 2: Design and Development of a Smart Mirror Using ESP32**Authors:** P. Kadam, S. Kadam**Publish Year:** 2019

About: This review examines the design and evolution of smart mirror systems utilizing ESP32 microcontrollers to integrate both home automation and primary security functions. The authors highlight the seamless combination of sensors, LED indicators, and camera modules to enhance situational awareness. The proposed system enables basic home monitoring, user notifications, and energy-efficient operation. A major contribution of the paper is demonstrating how ESP32 simplifies multi-function device development.

TITLE 3: Home Automation and Security Using ESP32-CAM Smart Mirror**Authors:** M. U. S. Kumar et al**.Publish Year:** 2020

About: This research presents a smart mirror integrated with ESP32-CAM, enabling both home automation tasks and security monitoring. The system captures images upon movement, sends notifications to the user, and allows remote access through IoT platforms

The authors demonstrate how ESP32-CAM enhances imaging quality and provides a responsive alert system suitable for indoor security. The study also explores customizable alert behavior based on environmental conditions. However, the system faces scalability limitations because the PIR sensor and camera range are insufficient for large homes, indicating that additional IoT nodes are needed for wider coverage.

TITLE 4: ESP32-Based Home Security and Monitoring System

Authors: R. A. A. S. Ranathunga et al

.Publish Year: 2021

About: This paper proposes a simple yet effective home security solution using ESP32 integrated with a PIR sensor and camera. The system transmits alerts and captured images to the user via Wi-Fi, offering remote monitoring capabilities. The authors highlight the ESP32's suitability for low-cost systems and its ability to support rapid detection with minimal delay. The research also discusses user interface integration for better control and alert handling. However, the study identifies limited compatibility with other advanced security components and emphasizes the need for stronger multi-sensor integration for improved reliability.

TITLE 5: IoT-Enabled Smart Mirror System for Home Security

Author: J. Sharma, D. Bansal, K. S. Bhardwaj

Publish Year: 2022

About: This paper introduces an IoT-enabled smart mirror equipped with ESP32, offering motion detection, instant notifications, and remote access via internet-based platforms. The system utilizes cloud storage for data management and supports real-time user updates, enhancing the overall responsiveness of home security. The authors highlight the benefits of IoT integration, including remote surveillance and continuous monitoring. Although effective, the system heavily relies on internet connectivity, making it vulnerable to network outages and delays. The study concludes that incorporating onboard intelligence (edge processing) could significantly improve security and reduce dependency on cloud services.

CHAPTER IV

METHODOLOGY

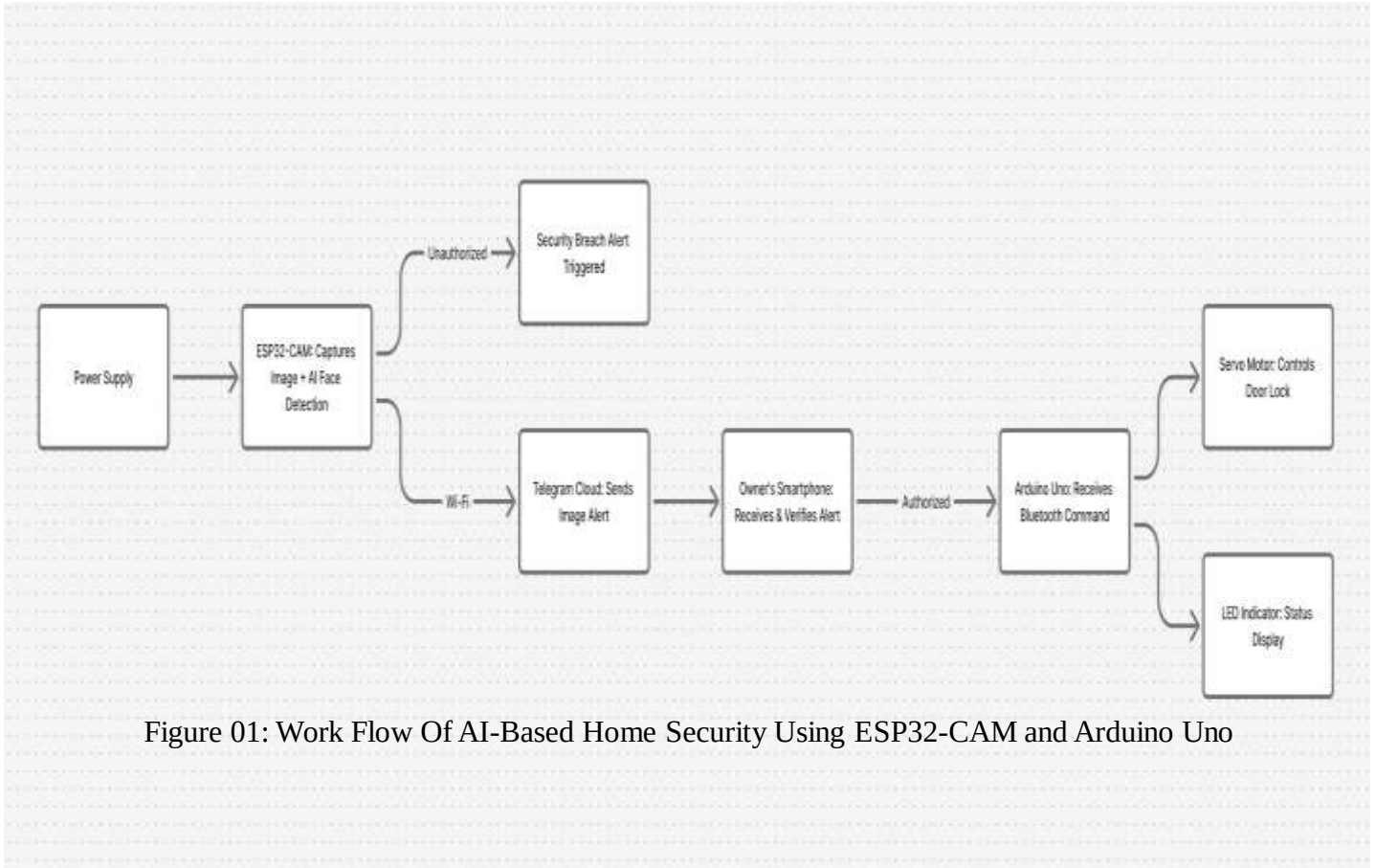


Figure 01: Work Flow Of AI-Based Home Security Using ESP32-CAM and Arduino Uno

The block diagram explains the working flow of the IoT-based home security system using the ESP32-CAM module. The ESP32-CAM captures the image of any person standing near the door and performs face recognition to identify whether the person is known or unknown. This forms the main detection and verification process of the system. The camera operates continuously in real time, ensuring that every visitor is monitored without manual intervention.

If the system identifies the person as unknown, an alert message along with the captured image is immediately sent to the homeowner through Telegram. This real-time notification helps the owner stay aware of any suspicious activity and respond quickly from anywhere.

After receiving the alert, the final door-access step is controlled manually by the homeowner using a Bluetooth mobile application. The app sends commands to a servo motor connected to the door lock, allowing the owner to open or close the door through simple ON/OFF switch controls. This ensures that only the owner has complete authority to grant access.

CHAPTER V

DATA DESCRIPTION

Software Components :

Micro Python libraries are as follows:

1. NumPy (np):

NumPy is used for fast numerical operations and efficient handling of image data arrays. It helps in processing and analyzing frames captured from the ESP32-CAM.

2. PIL (Image)

PIL allows opening, converting, and manipulating images received from the camera stream. It converts raw bytes into usable image formats for further processing.

3. requests

The requests library fetches the latest image from the ESP32-CAM using an HTTP link. It enables smooth communication between the Python script and the camera server.

Hardware Components :

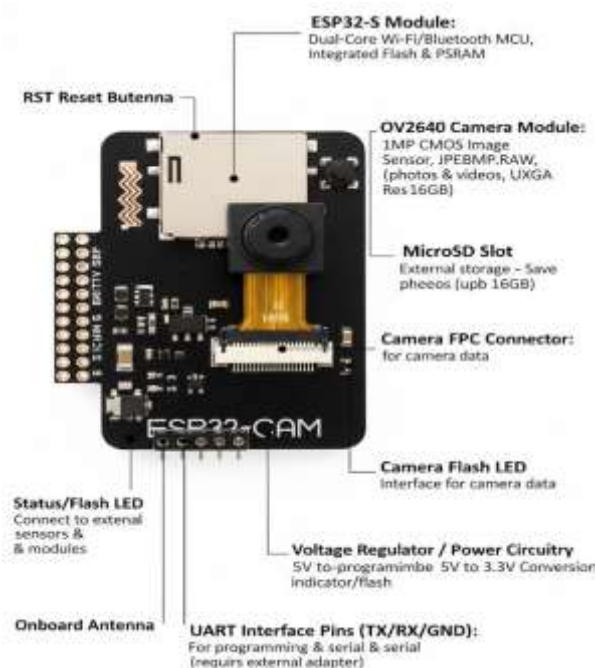


Figure 02: ESP32 CAM Module

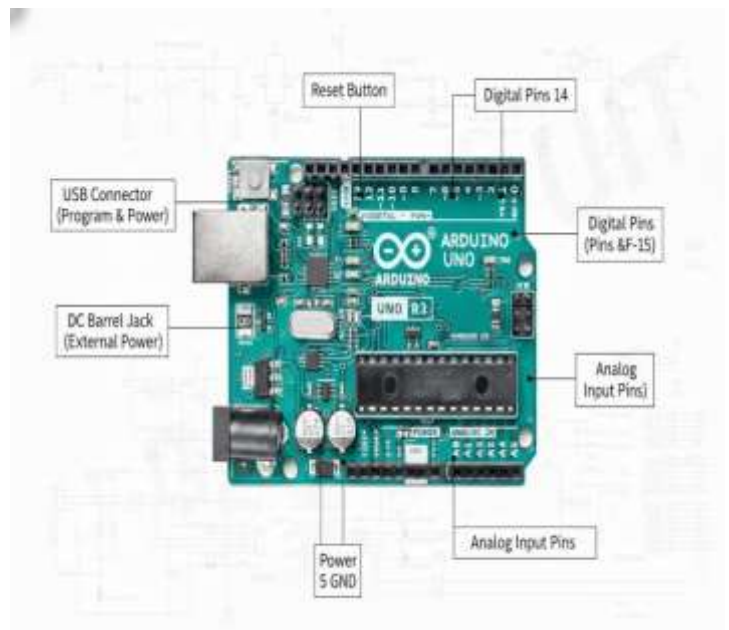


Figure 03: Arduino UNO Module

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ESP32-CAM captures intruder images and sends them via Telegram. Demonstrates reliable remote alerts and real-time monitoring.

2. Smart Wi-Fi Door Lock System Using ESP32-CAM and the Blynk App

J. P. Rohit, K. K. Naveen Kumar (2024).

Integrates ESP32-CAM and Blynk for IoT-based door locking. Enables camera monitoring and remote access control via smartphone.

3. Wi-Fi Door Lock System Using ESP32-CAM Based on IoT

S. Manikandan, M. Ramesh (2021).

Demonstrates door control via face recognition and IoT cloud. Highlights secure and user-friendly access management.

4. Home Door Security System with Face Recognition Using ESP32-CAM

S. S. Babu, R. K. Reddy et al. (2024).

Implements AI-based facial recognition for door access. ESP32-CAM sends alerts and captured images for remote verification.

5. Smart Home Warning System Using ESP32-CAM

P. Palkar, P. Bangar, S. Jadhav (2023).

ESP32-CAM detects motion and sends real-time alerts with images. Provides low-cost IoT surveillance without cloud dependency.